

Computer Application

Week 5 (Summer- 2023)

Networking Devices

1. Network Cables

Network cables are the transmission media to transfer data from one device to another. A commonly used network cable is category 5 cable with RJ – 45 connector, as shown in the image below:



2. Routers

A router is a transfers data packets between different computer networks. Typically, they are used to connect a PC or an organization's LAN to a broadband internet connection. They contain RJ-45 ports so that computers and other devices can connect with them using network cables.



3. Repeater, Hubs, and Switches

Repeaters, hubs and switches connect network devices together so that they can function as a single segment.

A repeater receives a signal and regenerates it before re-transmitting so that it can travel longer distances.

A hub is a multiport repeater having several input/output ports, so that input at any port is available at every other port.

A switch receives data from a port, uses packet switching to resolve the destination device and then forwards the data to the particular destination, rather than broadcasting it as a hub.



4. Bridges

A bridge connects two separate Ethernet network segments. It forwards packets from the source network to the destined network.



5. Gateways

A gateway connects entirely different networks that work upon different protocols. It is the entry and the exit point of a network and controls access to other networks.



6. Network Interface Cards

NIC is a component of the computer to connect it to a network. Network cards are of two types: Internal network cards and external network cards.



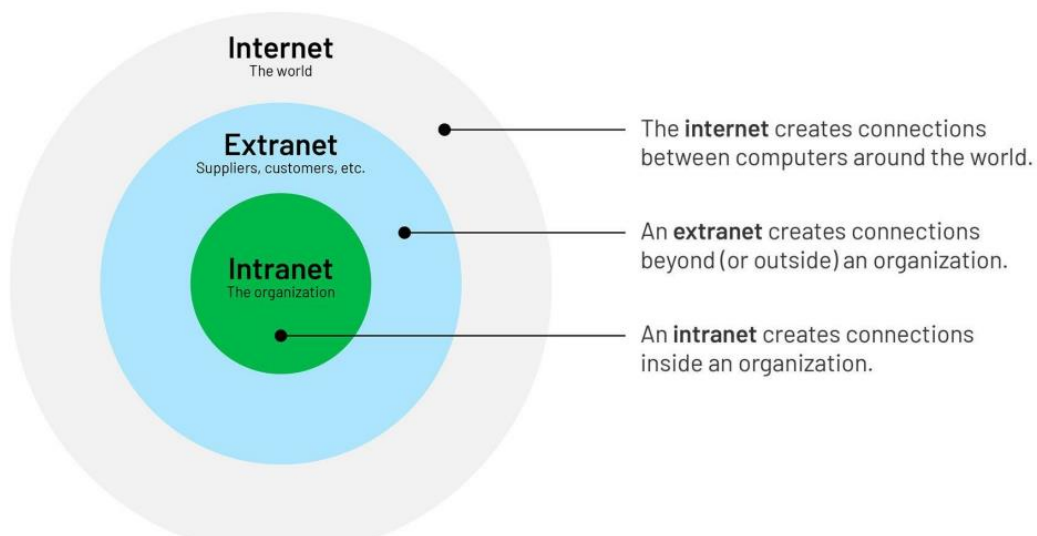
Internetworking

- ❖ Internetworking is the practice of inter-connecting multiple computer networks.
- ❖ The resulting system of inter-connected networks are called an inter-network, or simply an *internet*.
- ❖ The challenge when connecting various systems is to support communication between disparate technologies.

ARPANET

The U.S. Advanced Research Projects Agency Network (ARPANET) was the first public packet-switched computer network. It was first used in 1969 and finally decommissioned in 1989. ARPANET's main use was for academic and research purposes

The **Internet** is the global system of interconnected [computer networks](#) that uses the [Internet protocol suite](#) (TCP/IP) to communicate between networks and devices.



Parameter	Intranet	Extranet
Usage	Private	Private
User Types	Organization employees and Internal company departments	Suppliers, customer and Business partners.
Usage	Internal employee communication , telephone directories etc.	Check status of orders, Access data , send email
Security	High security. Configured under 100 security level in firewall	Generally uses VPN technology for secured communication over Internet. Medium security Level.
Regulated by	It is regulated by an organization.	It is regulated by multiple organization.
Ownership	Owned by Single organization	It is owned by single/multiple organization.

Network Model to Establish Communication

What Is the OSI Model?

The Open Systems Interconnection (OSI) model describes seven layers that computer systems use to communicate over a network. It was the first standard model for network communications, adopted by all major computer and telecommunication companies in the early 1980s

The modern Internet is not based on OSI, but on the simpler TCP/IP model. However, the OSI 7-layer model is still widely used, as it helps visualize and communicate how networks operate, and helps isolate and troubleshoot networking problems.

OSI was introduced in 1983 by representatives of the major computer and telecom companies, and was adopted by ISO as an international standard in 1984.

OSI Model Explained: The OSI 7 Layers

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

We'll describe OSI layers “top down” from the application layer that directly serves the end user, down to the physical layer.

7. Application Layer

The application layer is used by end-user software such as web browsers and email clients. It provides protocols that allow software to send and receive information and present meaningful data to users. A few examples of application layer protocols are the [Hypertext Transfer Protocol](#) (HTTP), File Transfer Protocol (FTP), Post Office Protocol (POP), Simple Mail Transfer Protocol (SMTP), and Domain Name System (DNS).

6. Presentation Layer

The presentation layer prepares data for the application layer. It defines how two devices should encode, encrypt, and compress data so it is received correctly on the other end. The presentation layer takes any data transmitted by the application layer and prepares it for transmission over the session layer.

5. Session Layer

The session layer creates communication channels, called sessions, between devices. It is responsible for opening sessions, ensuring they remain open and functional while data is being transferred, and closing them when communication ends. The session layer can also set checkpoints during a data transfer—if the session is interrupted, devices can resume data transfer from the last checkpoint.

4. Transport Layer

The transport layer takes data transferred in the session layer and breaks it into “segments” on the transmitting end. It is responsible for reassembling the segments on the receiving end, turning it back into data that can be used by the session layer. The transport layer carries out flow control, sending data at a rate that matches the connection speed of the receiving device, and error control, checking if data was received incorrectly and if not, requesting it again.

3. Network Layer

The network layer has two main functions. One is breaking up segments into network packets, and reassembling the packets on the receiving end. The other is routing packets by discovering the best path across a physical network. The network layer uses network addresses (typically Internet Protocol addresses) to route packets to a destination node.

2. Data Link Layer

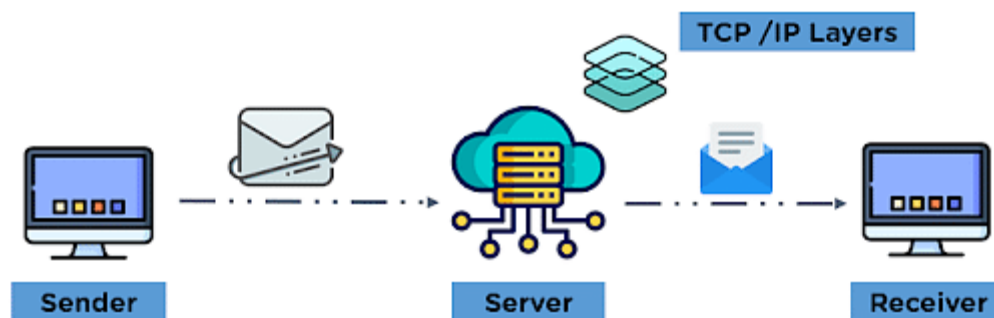
The data link layer establishes and terminates a connection between two physically-connected nodes on a network. It breaks up packets into frames and sends them from

source to destination. This layer is composed of two parts—Logical Link Control (LLC), which identifies network protocols, performs error checking and synchronizes frames, and Media [Access Control](#) (MAC) which uses MAC addresses to connect devices and define permissions to transmit and receive data.

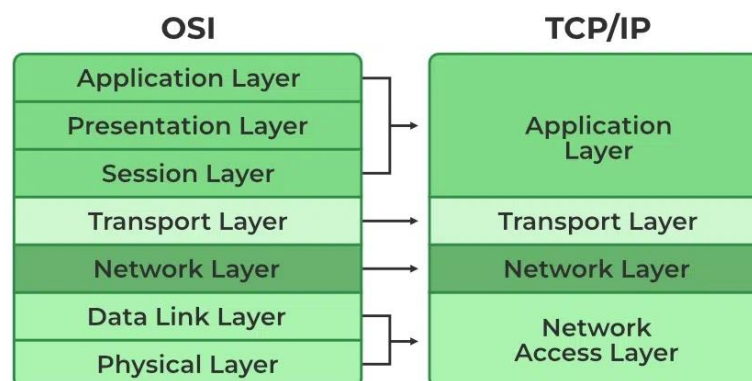
1. Physical Layer

The physical layer is responsible for the physical cable or wireless connection between network nodes. It defines the connector, the electrical cable or wireless technology connecting the devices, and is responsible for transmission of the raw data, which is simply a series of 0s and 1s, while taking care of bit rate control.

What is TCP/IP Model?



The TCP/IP model refers to the Transmission Control Protocol/[Internet Protocol](#) Model. This model is a part of the network domain designed specifically for overseeing efficient and error-free transmission of data.



OSI vs. TCP/IP Model

Parameters	OSI Model	TCP/IP Model
No. of Layers	There are 7 layers.	There are 4 layers.
Acronyms	OSI stands for open system interconnection.	TCP/IP stands for transmission control protocol/internet protocol
Developed by	ISO	Department of Defense (DoD)
Layer Separation	OSI model has a separate Presentation layer and Session layer.	TCP/IP does not have a separate Presentation layer or Session layer.
Protocol implementation	Model was defined before implementation takes place.	Model defines after protocol were implemented.
Model Concept	based on three concept i.e. Service, interface and protocol.	It did not distinguish between service, interface and protocol.
Reliable delivery	It gives guarantee of reliable delivery of packet.	It does not give guarantee of reliable delivery of packet.